

CET138 What is Full Stack Development?

Introduction

Full Stack Development refers to the ability of a developer to work on both the frontend and backend of a web application. A full stack developer has expertise across the entire technology stack from the user interface that visitors see in their browser to the server and database that power the application behind the scenes.

The term stack describes the combination of technologies and frameworks used in a web application. A typical stack includes:

- Frontend: The client-side technologies (HTML, CSS, JavaScript)
- Backend: The server-side technologies (databases, server frameworks, APIs)
- DevOps: Deployment, hosting, and infrastructure management

Full stack developers understand how these layers interact with each other, allowing them to build complete, functional web applications independently or as part of a team.

Code Snippet Navbar Structure

```
<nav class="navbar navbar-expand-lg fixed-top" id="mainNav">
  <div class="container">
    <a class="navbar-brand nav-brand" href="index.html">Rajesh Karki</a>
    <ul class="navbar-nav mx-auto gap-1">
      <li class="nav-item"><a class="nav-link" href="index.html">Home</a></li>
      <li class="nav-item"><a class="nav-link" href="about.html">Full Stack</a></li>
      <li class="nav-item"><a class="nav-link" href="html.html">HTML</a></li>
      <li class="nav-item"><a class="nav-link" href="css.html">CSS</a></li>
      <li class="nav-item"><a class="nav-link" href="bootstrap.html">Bootstrap</a></li>
      <li class="nav-item"><a class="nav-link" href="javascript.html">JavaScript</a></li>
    </ul>
  </div>
</nav>
```

Explanation of Code

- `<nav>` element: Semantic HTML tag that defines the navigation section of the page.
- `navbar navbar-expand-lg`: Bootstrap classes that create a responsive navigation bar.
- `fixed-top`: Keeps the navbar fixed at the top of the viewport while the user scrolls.
- `.navbar-brand`: Creates a branded logo or site name in the navbar.
- `navbar-nav mx-auto`: Bootstrap utility classes that center the navigation links.
- `href` attributes: Link to different pages, demonstrating frontend separation.

Demonstration of Understanding

- Frontend: Using HTML, CSS, and JavaScript to create an interactive user interface.
- Styling: Implementing Bootstrap for consistent, professional design across all pages.
- Interactivity: JavaScript handles theme toggling, form submissions, and DOM manipulation.
- Multi-page structure: Separate HTML pages for each topic with dedicated content.
- Integration: All components work seamlessly together across the full stack.

Conclusion

Full Stack Development is not just about knowing multiple languages but understanding how all components of a web application work together. This ePortfolio demonstrates a complete, functional web application that showcases frontend technologies combined with responsive design principles using Bootstrap.